Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building C++/Python systems, perception-oriented ML pipelines, backend tooling, and automation workflows across simulation, computer vision, and distributed engineering environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a **GPT-based Decision Transformer** for offline RL in autonomous driving, improving long-horizon trajectory planning by **18%** in simulation while developing end-to-end Python pipelines across 100K+ trajectory logs.
- Architected a **spatial-temporal encoder** combining HD-map geometry, lane direction, and ego-agent history through multi-head attention, improving closed-loop planning stability by **15%** for autonomous-driving evaluation, simulation, and validation workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and multimodal workflows** handling 10,000+ daily requests, integrating OCR, file ingestion, and model-backed evaluation pipelines that strengthened perception-style processing, reliability, and tooling for production users.
- Built and debugged **offline analysis and validation utilities** for uploads, OCR, persistence, and grading flows, reducing feedback turnaround by **60%** while improving observability, reproducibility, and issue isolation across model workflows.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and data workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, schema updates, and maintainable release pipelines across cloud-hosted systems and internal tooling.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and production handoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, automation workflows, and reliable release coordination across operating environments.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions across Windows, Linux, and macOS environments during recurring build and test cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows while strengthening high-volume reliability, service coordination, and issue resolution for active users.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and team handoffs during debugging and production-readiness checks.

Skills

Languages: **C++**, **Python**, Java, JavaScript, TypeScript, SQL, Bash
Robotics/ML: PyTorch, Transformers, offline RL, computer vision, trajectory planning, map-aware modeling, evaluation pipelines
Systems: Linux/Unix, object-oriented design, data structures, algorithms, CI/CD, debugging, automation workflows
Tools: Git/GitHub, Jenkins, Docker, REST APIs, WebSockets, AWS Glue/S3, Azure SQL, PostgreSQL, data visualization, offline analysis

Projects

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81%** to **96%** through repeatable evaluation pipelines and structured error analysis.

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*
- Built an **AI-powered image platform** using CLIP, SAM, Stable Diffusion, and MongoDB to support segmentation, retrieval, recoloring, and restoration workflows with scalable storage, subscriptions, and production-style feature orchestration.

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads, strengthening systems reasoning and correctness-focused testing practices.